Generated using the official AMS LaTeX template v5.0

On Robust Diagnostic Planetary Geostrophy

Thomas W. N. Haine*

Earth & Planetary Sciences, Johns Hopkins University, Baltimore, MD,

Extra Author

Affiliation, City, State/Province, Country

Corresponding author: Thomas Haine, Thomas.Haine@jhu.edu

ABSTRACT

Enter the text of your abstract here. This is a sample American Meteorological Society (AMS) LaTeX template. This document provides authors with instructions on the use of the AMS LaTeX template. Authors should refer to the file amspaper.tex to review the actual LaTeX code used to create this document. The template.tex file should be modified by authors for their own manuscript.

1. Introduction

The large-scale, low-frequency ocean general circulation is controlled by hydrostatic and geostrophic dynamics (e.g., see Pedlosky 1987, Section 8.8 of Salmon 1998 and Section 5.3 of Vallis 2006). The flow is forced by (i) surface stress from the atmosphere, and (ii) air/sea fluxes of heat and freshwater that determine the seawater temperature and salinity, and thereby its density (equivalently, its buoyancy, which is a negative scaled density). The flow is dissipated by viscous friction and property mixing that act on small-scales. The density field and the velocity field are coupled. These dynamics are called *planetary geostrophic* (PG), or sometimes the *thermocline equations* (Robinson and Stommel 1959; Welander 1959; de Verdière 1988).

Understanding how the general circulation, and the large-scale density field, depend on air/sea forcing and small-scale dissipation is a central goal of physical oceanography. For instance, we want to quantify the processes responsible for the gross structure of the thermocline and the associated field of motion. Achieving this goal has been difficult, however, for several reasons. First, PG dynamics are nonlinear. The nonlinearity does not arise from inertia (momentum advection) or the seawater equation of state, although these are both prominent features of more complete dynamical models. Instead, the nonlinearity arises from the advection of density. Second, PG dynamics are non-conservative. That is, they describe a forced-dissipative balance without preserved invariants, like energy and potential vorticity. Third, PG dynamics are non-local. In particular, the pressure at a particular point depends on remote properties. This is most obviously seen in the hydrostatic balance, which equates the pressure with the integrated weight of the water above the point in question. Fourth, the density and velocity fields are coupled in a nearly inextricable way for the (realistic) case of variable ocean depth. For these reasons, the challenge of understanding the general circulation has also been called the *thermocline problem* (e.g., Veronis 1969; Welander

1971). Each of these reasons presents technical obstacles to theoretical progress on solving PG dynamics.

Much prior work has focused on linear PG dynamics, with a rich theoretical literature stretching back to the early twentieth century (for a textbook presentation, see chapter 3 of Salmon 1998, and for an historical essay, see Pedlosky 2006). Seminal milestones in this literature include the following: The mechanism of stress transmission via thin, unstratified viscous-geostrophic boundary layers and inviscid interior vortex stretching by pumping or suction was pioneered by Ekman (1905) (see also Greenspan 1990). The origin of western intensification of subtropical gyre flow was elucidated by the flat-bottomed, unstratified models of Stommel (1948) and Munk (1950), which pointed out the essential effect of meridionally-varying Coriolis parameter f. Stommel and Arons (1959) applied similar ideas to western intensification of abyssal flow fed by a point source of mass. Stratification in its simplest form was introduced in layer models, such as that by Welander (1968).

Three conceptual insights were achieved around this time: First, the guiding role of f/H geostrophic contours was recognized, where H is the ocean depth. For example, unstratified, unforced, and undamped PG flow must conserve f/H. Therefore, streamlines coincide with f/H contours (Welander 1968). Stratification can shield the upper circulation from this topographic constraint (Anderson and Gill 1975; Rhines and Young 1982; Marshall 1995; Spall 2016). In the presence of surface Ekman pumping/suction and linear bottom drag, the equation for the streamfunction of the horizontal flow is isomorphic to a forced advection-diffusion equation. In other words, the streamfunction is analogous a passive scalar tracer field advected by f/H streamlines, diffused by bottom drag, and forced by the wind-driven surface Ekman layer (Welander 1968). The so-called *β plume* is a canonical example in which a plume of streamfunction contours

spreads westward from an isolated Ekman pumping source on an unstratified, flat-bottomed, β-plane ($\beta = df/dy$ measures the variation in the Coriolis parameter with latitude y; Stommel 1982).

Notable twenty-first century advances include the following. Nøst and Isachsen (2003) exploit the alignment of streamlines and f/H contours to infer the circulation given the surface stress and the density field (see also Aaboe and Nøst 2008; Nilsson et al. 2024). To do so, they assume simple forms for the bottom drag and use climatological density data: that is, they exploit observations of the temperature and salinity fields and do not try to infer them. The circulation on a particular closed f/H contour is computed from the density data and the integrated surface stress curl over the area enclosed by the f/H contour (or, equivalently, the total torque of the stress around the contour). Papers in the last few years have explored the nature and consequences of the boundary layer at the sea floor (e.g., Peterson and Callies 2023, 2026). Diapycnal (near vertical) mixing is known to be enhanced near the bottom. Also, the heat and freshwater fluxes (hence the buoyancy flux) across the sea floor are very small. This constraint implies a non-zero horizontal density gradient near sloping topography that associates with horizontal pressure gradients and therefore geostrophic flow. These bottom boundary layer processes therefore provide a mechanism to induce circulation, in addition to air/sea stress and buoyancy exchange.

To our knowledge, no theoretical treatment exists of the linear, stratified, PG problem with variable topography and Coriolis parameter, plus accurate surface and bottom boundary layers. The central question tackled in this paper is therefore: Given a three-dimensional density field, and a surface stress field, what is the PG flow? Equivalently: How does the PG flow at a particular point depend on the buoyancy field and the surface stress at other points? We answer this question for an arbitrary basin shape, a variable Coriolis parameter, and resolved surface and bottom boundary layers. The deep tropics, where the Coriolis parameter vanishes and PG breaks down, are excluded. An analytic solution is provided, not a numerical model, although, in general, a two-dimensional

elliptic differential equation for the surface pressure (and thus the surface height) field must be solved numerically. Software to derive and compute solutions is provided too. There are several benefits, including: conceptual transparency, access to parameter regimes that are expensive (or impossible) to reach numerically, and speed of calculation. There are some disadvantages, such as limitations on the forms of viscous and diabatic dissipation. Even for these cases, the approach described here is enlightening.

Answering this question is a step towards the thermocline problem. It provides new physical and mathematical insight and a framework that may be useful to tackle the broader question. Avoiding density advection amounts to, essentially, specifying the density field. This density specification has been called a *robust diagnostic* assumption (Sarmiento and Bryan 1982; see also Bogden et al. 1993; Park and Guernier 2001; Nøst and Isachsen 2003; Nilsson et al. 2005). Note that considering density (or buoyancy) alone is not a restriction if one wants to compute temperature and salinity individually.

The layout of the rest of the paper is as follows: Section 2 explains the robust diagnostic PG equations and their nondimensionalization. Section 3 lays out several diagnostic dynamical balances that apply to the depth-integrated horizontal current and its vorticity. In Section 4 the linear buoyancy equation is solved given a relaxation buoyancy field and a surface buoyancy boundary condition. This step prototypes the main task of solving the PG equations, which is covered in Section 5. Section 5 includes the definition of the elliptic equation for the surface pressure field. Section 6 exemplifies the method with case studies. Finally, Section 7 is a discussion of the utility of the PG solution with comments on future applications and research opportunities. The Appendices contain supporting material that is mathematical or numerical.

2. Governing Equations

a. Robust Diagnostic Planetary Geostrophy

The horizontal flow (u, v) satisfies the dimensional PG equations written in Cartesian coordinates (x, y, z):

$$-fv = -(1/\rho_0)\partial_x p + \partial_z\left(\nu \partial_z u\right), \tag{1}$$

$$fu = -(1/\rho_0)\partial_y p + \partial_z\left(\nu \partial_z v\right), \tag{2}$$

(see Eqs. (1)–(2) of Peterson and Callies 2026, Eq. (3.26) of Klinger and Haine 2019, and Eq. (2.274) of Vallis 2006). The pressure is p, the characteristic seawater density is ρ_0, the vertical viscosity is ν, and the Coriolis parameter is $f(y)$. These equations presume Boussinesq, time-independent, and linear (small Rossby number) dynamics. They apply to the ocean general circulation on horizontal scales larger than the first baroclinic deformation radius (hundreds of kilometers and larger). Dissipation is included with the vertical friction terms, but there is no horizontal friction (see Section 7). As vertical friction is included, the solutions to (1) and (2) resolve the surface and bottom Ekman layers. The boundary conditions therefore specify a known stress (from the wind) at the surface and no flow at the bottom:

$$\text{Surface } z = \eta : \nu\rho_0\, \partial_z(u, v) = (\tau^x, \tau^y), \tag{3}$$

$$\text{Bottom } z = -H : (u, v) = (0, 0), \tag{4}$$

where $H(x, y)$ is the variable bottom depth (and H is positive so that z measures height upwards from the average sea level at $z = 0$), and η is the height of the sea surface.

The fluid is incompressible so that the three-dimensional flow $\mathbf{u} = (u, v, w)$ is non-divergent:

$$\nabla \cdot \mathbf{u} = 0. \tag{5}$$

The sea surface and the sea floor are impermeable so that the normal component of the velocity vanishes, $\mathbf{u} \cdot \hat{\mathbf{n}} = 0$ with $\hat{\mathbf{n}}$ the unit outward pointing normal vector at the boundary. The absolute buoyancy is defined as a scaled density

$$b_{\text{abs}} = -\frac{g}{\rho_0}\rho \equiv -g - \frac{g(\rho - \rho_0)}{\rho_0} \equiv b - g, \tag{6}$$

where g is the uniform gravitational acceleration. The buoyancy anomaly b is the main variable of interest (hereafter just called the buoyancy). The absolute pressure field satisfies hydrostatic balance, which is:

$$\begin{aligned} &\partial_z p_{\text{abs}} = \rho_0 b_{\text{abs}}, \\ \implies &\partial_z (p_0 + p) = \rho_0 (b - g), \\ &\text{with } \partial_z p_0 \equiv -\rho_0 g \text{ (and } p_0(z = 0) = 0), \\ \implies &\partial_z p = \rho_0 b. \end{aligned} \tag{7}$$

The dynamic pressure anomaly p is the main variable of interest (hereafter just called the pressure). Finally, the buoyancy field satisfies the time-independent advection-diffusion-relaxation equation

$$\mathbf{u} \cdot \nabla b = \partial_z (\kappa \partial_z b) + \gamma (B - b). \tag{8}$$

The vertical mixing of buoyancy depends on the diffusivity coefficient κ (and there is no horizontal buoyancy mixing, consistent with the momentum equations). The final term in (8) is relaxation to a known background buoyancy field $B(x, y, z)$ at a rate γ. This is the robust diagnostic assumption. The boundary conditions on buoyancy are:

$$\text{Surface } z = \eta : b = b_s \tag{9}$$

$$\text{Bottom } z = -H : \nabla b \cdot \hat{\mathbf{n}} = 0, \tag{10}$$

where $b_s(x, y)$ is a known surface buoyancy field and at the sea floor $\hat{\mathbf{n}} = -(\partial_x H, \partial_y H, 1)/\sqrt{1 + (\partial_x H)^2 + (\partial_y H)^2}$. Notice that the bottom boundary condition is no nor-

mal flux of buoyancy through the sea floor. At a sloping bottom boundary, this condition will lead to non-zero horizontal buoyancy gradients, and therefore a non-zero thermal wind (Peterson and Callies 2026; see, for example, Klinger and Haine 2019 Sections 2.1.2 and 2.2.2 for the thermal wind, which is derived from the geostrophic and hydrostatic balances). We retain both inhomogeneous sources of buoyancy, b_s and B, for generality, although both of them are not always needed.

The viscosity and diffusivity parameters in (1)–(2) and (8) can vary in the vertical direction, for instance, to represent the effects of bottom-enhanced mixing (see Section 7, Peterson and Callies 2023, 2026). Most attention is on the case in which these coefficients are uniform, however.

b. Non-dimensionalization

For convenience, the equations in Section 2a are non-dimensionalized with the following scales:

$$
\begin{aligned}
(x, y) \sim L,\ z \sim H_0,\ t \sim L/U_0,\\
(u, v) \sim U_0,\ w \sim (H_0/L)U_0,\\
p \sim \rho_0 f_0 L U_0,\ b \sim {N_0}^2 H_0,\\
f \sim f_0,\ \kappa \sim \kappa_0,\ \nu \sim \nu_0,\ (\tau^x, \tau^y) \sim \tau_0
\end{aligned}
$$

where $L, H_0, f_0, N_0, \kappa_0$, and ν_0 are the characteristic values of the ocean basin width, ocean depth, Coriolis parameter, buoyancy frequency (from B or b_s), diffusivity, and viscosity, respectively. The characteristic horizontal current speed based on the thermal wind relation is $U_0 = (N_0^2 H_0^2)/(f_0 L)$. The value of the stress scale τ_0 is specified below.

With these scalings, the system of governing equations (1)–(10) in non-dimensional form becomes:

$$-fv = -\partial_x p + \mathrm{Ek}\,\partial_z\left(\nu\partial_z u\right), \tag{11}$$

$$fu = -\partial_y p + \mathrm{Ek}\,\partial_z\left(\nu\partial_z v\right), \tag{12}$$

$$\partial_x u + \partial_y v + \partial_z w = 0, \tag{13}$$

$$\partial_z p = b, \tag{14}$$

$$\mathrm{Pr}\,\mathrm{Bu}\left(u\partial_x b + v\partial_y b + w\partial_z b\right) = \mathrm{Ek}\,\partial_z\left(\kappa\partial_z b\right) + \Gamma\left(B - b\right) \tag{15}$$

(where the variables u, v, w, b, and p now mean the non-dimensional fields, and so on). The boundary conditions are

$$\left[\nu\,\partial_z(u,v)\right]_{z=\eta} = (\tau^x, \tau^y), \tag{16}$$

$$b|_{z=\eta} = b_s, \tag{17}$$

at the surface, and

$$(u,v)|_{z=-H} = (0,0), \tag{18}$$

$$\left[\delta^2\left(\partial_x b\,\partial_x H + \partial_y b\,\partial_y H\right) + \partial_z b\right]_{z=-H} = 0 \tag{19}$$

at the bottom. The upper stress boundary condition provides the scaling $\tau_0 = \nu_0\rho_0 U_0/H_0$, and the lower buoyancy boundary condition requires the domain aspect ratio $\delta = H_0/L$. The aspect ratio δ is assumed to be small everywhere, $\delta \ll 1$ (which implies that vertical side walls are excluded).

This scaling process introduces five non-dimensional parameters:

$$\mathrm{Ek} = \nu_0/(f_0 H_0^2), \quad \text{Ekman number,} \tag{20}$$

$$\mathrm{Bu} = N_0^2 H_0^2/(f_0^2 L^2), \quad \text{Burger number,} \tag{21}$$

$$\mathrm{Pr} = \nu_0/\kappa_0, \quad \text{Prandtl number,} \tag{22}$$

$$\Gamma = \mathrm{Pr}\,\gamma/f_0 \quad \text{relaxation number,} \tag{23}$$

$$\delta = H_0/L \quad \text{aspect ratio.} \tag{24}$$

This system is closely related to equations (9)–(13) of Peterson and Callies (2026) although there are some minor differences in detail.

3. Diagnostic Dynamical Balances

Several important dynamical balances are derived from the non-dimensionalized governing equations (11)–(19). They involve manipulations of the momentum equations such as integrating vertically and forming a vorticity equation by taking the curl. In general, these manipulations do not commute, so their order matters and a variety of distinct dynamical balances emerges. The goal is to isolate and elucidate the link between key circulation properties, like the depth-averaged horizontal current, and forcing fields, like the surface stress and the buoyancy field. In what follows, Waldman and Giordani (2023) is a useful guide and covers a more general set of governing equations than (11)–(19). Table 1 summarizes salient properties of the diagnostic dynamical balances.

a. Vertically-Integrated Flow

Consider first the vertically-integrated horizontal flow (U,V). It is defined by:

$$U \equiv \int_{-H}^{0} u \, dz = -\int_{-H}^{0} z\,\partial_z u \, dz,$$
$$V \equiv \int_{-H}^{0} v \, dz = -\int_{-H}^{0} z\,\partial_z v \, dz, \tag{25}$$

where we exploit the bottom boundary condition (18) on (u,v). Consistent with the linearized small-surface-displacement approximation, $|\eta| \ll H$, used below, the upper limit of vertical integration is evaluated at $z = 0$ (not $z = \eta$). The equations for (U,V) come from vertically integrating the three-dimensional momentum (3DM) equations (11), (12):

$$-fV = -\int_{-H}^{0} \partial_x p \, dz + \mathrm{Ek}\int_{-H}^{0} \partial_z\left(\nu\partial_z u\right) \, dz, \tag{26}$$
$$fU = -\int_{-H}^{0} \partial_y p \, dz + \mathrm{Ek}\int_{-H}^{0} \partial_z\left(\nu\partial_z v\right) \, dz. \tag{27}$$

The depth-integrated pressure gradient is, for example, in the x direction:

$$\int_{-H}^{0} \partial_x p \, dz = -p_b\,\partial_x H + \partial_x \int_{-H}^{0} p \, dz, \tag{28}$$
$$= -p_b\,\partial_x H + \partial_x\left[p_b H - \int_{-H}^{0} z\,\partial_z p \, dz\right], \tag{29}$$
$$= H\,\partial_x p_b - \partial_x \int_{-H}^{0} zb \, dz, \tag{30}$$

using the Leibniz integration formula,[1] where $p_b = p(x, y, -H)$ is the bottom pressure, integration by parts,[2] and the hydrostatic relation (14). The viscous term in the x direction is:

$$\int_{-H}^{0} \partial_z \left(\nu \, \partial_z u \right) \, dz = \left[\nu \, \partial_z u \right]_{-H}^{0} = \tau^x - \nu \, \partial_z u |_{z=-H} \,, \tag{31}$$

which uses the leading order form of the boundary condition (16), evaluated at $z = 0$.

Therefore, (26) and (27) lead to the (time-independent, linear) depth-integrated momentum (DIM, also called the barotropic momentum) equations:

$$-fV = -H \, \partial_x p_b - \partial_x \chi + \mathrm{Ek} \left(\tau^x + \tau_b^x \right) , \tag{32}$$

$$fU = -H \, \partial_y p_b - \partial_y \chi + \mathrm{Ek} \left(\tau^y + \tau_b^y \right) , \tag{33}$$

where

$$\chi = - \int_{-H}^{0} z b \, dz \tag{34}$$

is the baroclinic potential energy, and $-\nu \, \partial_z (u, v)|_{z=-H} \equiv (\tau_b^x, \tau_b^y) = \tau_b$ is the bottom stress on the fluid (see Hughes and Killworth 1995 Eq. (14) and Waldman and Giordani 2023 Eq. (9)). The DIM equations relate the depth-integrated horizontal flow (U, V) to the net stress, the gradient of bottom pressure p_b, and the gradient of baroclinic potential energy χ.

---

[1] The Leibniz integration formula reads:

$$\partial_x \left(\int_{a(x)}^{b(x)} f(x, y) \, dy \right) = \partial_x b f(x, b(x)) - \partial_x a f(x, a(x)) + \int_{a(x)}^{b(x)} \partial_x f(x, y) \, dy .$$

[2] Integration by parts gives:

$$\begin{aligned} \int_{-H}^{0} z \frac{\partial p}{\partial z} \, dz &= \left[z p \right]_{-H}^{0} - \int_{-H}^{0} p \, dz , \\ &= p_b H - \int_{-H}^{0} p \, dz . \end{aligned}$$

b. Vorticity Balances for the Depth-Integrated Flow

Apart from the understanding the balance of terms controlling U and V in the DIM equations, it is useful to understand the balance of vorticity for the depth-integrated circulation. Waldman and Giordani (2023) show how to derive four equations that represent different versions of these vorticity dynamics, and they are now described in turn.

1) Barotropic vorticity equation

The first option is to take the curl of the DIM equations ∂_x(33)-∂_y(32), which gives: [3]

$$\partial_x \{fU\} - \partial_y \{-fV\} = \partial_x \left\{-H\,\partial_y p_b - \partial_y \chi + \mathrm{Ek}\left(\tau^y + \tau_b^y\right)\right\} - \partial_y \left\{-H\,\partial_x p_b - \partial_x \chi + \mathrm{Ek}\left(\tau^x + \tau_b^x\right)\right\}$$

$$\implies f\left(\partial_x U + \partial_y V\right) + \beta V = J(p_b, H) + \mathrm{Ek}\,\hat{\mathbf{z}} \cdot [\nabla \times (\tau + \tau_b)], \tag{36}$$

where $J(a,b) \equiv \partial_x a\,\partial_y b - \partial_y a\,\partial_x b = (\nabla a \times \nabla b) \cdot \hat{\mathbf{z}}$ is the Jacobian operator, $\beta = df/dy$, and $\hat{\mathbf{z}}$ is the unit upward vector. This is the time-independent, linear equation for the vorticity of the depth-integrated flow, also called the barotropic vorticity (BV) equation (see Waldman and Giordani 2023 Eq. (10)). For the time-independent case, the horizontal divergence of the vertically-integrated flow (the first term on the left hand side of (36)) vanishes. This constraint comes from the non-divergence equation (13), which implies that

$$\int_{-H}^{0} \partial_x u + \partial_y v + \partial_z w \; dz = 0,$$

$$\implies \partial_x U + \partial_y V = 0. \tag{37}$$

Thus, for time-independent solutions:

$$J(p_b, H) + \mathrm{Ek}\,\hat{\mathbf{z}} \cdot [\nabla \times (\tau + \tau_b)] - \beta V = 0. \tag{38}$$

[3] Recall that

$$\hat{\mathbf{z}} \cdot (\nabla \times \mathbf{U}) = \partial_x V - \partial_y U. \tag{35}$$

For the flat-bottom case (in which the Jacobian term vanishes), this equation coincides with the vorticity equation for the so-called homogeneous model of the ocean circulation (Pedlosky 1996; Vallis 2006; Klinger and Haine 2019). It provides the basis for the classic subtropical gyre theories of Sverdrup (1947); Stommel (1948), and Munk (1950).

2) Depth-Integrated Vorticity equation

The second option is to switch the order of operations used to produce the BV equation, namely, form a depth-dependent vorticity equation first, then vertically integrate. Taking the curl of the 3DM equations ∂_x(12)-∂_y(11) gives

$$\partial_x (fu) + \partial_y (fv) = \mathrm{Ek}\, \partial_z (\nu \partial_z \zeta), \tag{39}$$

where $\zeta = \partial_x v - \partial_y u$ is the vertical component of the relative vorticity vector. Using the non-divergence constraint (13) implies that

$$\beta v - f \partial_z w = \mathrm{Ek}\, \partial_z (\nu \partial_z \zeta). \tag{40}$$

Integrating in the vertical gives the Sverdrup relation

$$\beta V + f w|_{z=-H} = \mathrm{Ek}\, \hat{\mathbf{z}} \cdot [\nabla \times (\tau + \tau_b)] \tag{41}$$

(Pedlosky 1996 Eq. (1.2.15); Waldman and Giordani 2023 Eq. (4)). This depth-integrated vorticity (DIV) equation is a balance between advection of planetary vorticity by the depth integrated flow, vortex stretching, and the net torque imposed by boundary stresses.

3) Geostrophic Contour Vorticity Balance

The third option is to write the equation for the vorticity of the depth-averaged flow. That is, divide the DIM equations (32)–(33) by water depth H and take the curl:

$$\partial_y\left(-\frac{fV}{H}\right)-\partial_x\left(\frac{fU}{H}\right)=\partial_y\left[-\partial_x p_b-\frac{1}{H}\partial_x\chi+\frac{\mathrm{Ek}}{H}\left(\tau^x+\tau_b^x\right)\right]-\partial_x\left[-\partial_y p_b-\frac{1}{H}\partial_y\chi+\frac{\mathrm{Ek}}{H}\left(\tau^y+\tau_b^y\right)\right]. \tag{42}$$

One finds that

$$\begin{aligned}\partial_y\left(-\frac{f}{H}\partial_x\Psi\right)+\partial_x\left(\frac{f}{H}\partial_y\Psi\right)&=-\partial_y\left(\frac{1}{H}\partial_x\chi\right)+\partial_x\left(\frac{1}{H}\partial_y\chi\right)-\mathrm{Ek}\,\hat{\mathbf{z}}\cdot\left(\nabla\times\frac{\tau+\tau_b}{H}\right),\\ \implies J\left(\frac{f}{H},\Psi\right)&=J\left(\frac{1}{H},\chi\right)-\mathrm{Ek}\,\hat{\mathbf{z}}\cdot\left(\nabla\times\frac{\tau+\tau_b}{H}\right),\end{aligned} \tag{43}$$

where the streamfunction of the depth-integrated horizontal flow Ψ satisfies

$$(U,V)=\left(-\partial_y\Psi,\partial_x\Psi\right). \tag{44}$$

This recipe (vertically integrate, divide by H, then take the curl) eliminates the bottom pressure. Equation (43) is called the (linear, time-independent) geostrophic contour vorticity balance (GCVB) by Waldman and Giordani (2023) (although Peterson and Callies 2026 call it the "barotropic vorticity equation"). The first term on the right hand side $J(1/H,\chi)$ is called the joint effect of baroclinicity and relief (JEBAR) term (Sarkisyan and Ivanov 1971).

The GCVB (43) is an equation for the streamfunction of the depth-integrated flow Ψ given the buoyancy field (and hence the JEBAR term) and the surface and bottom stresses. The GCVB shows how depth-integrated flow Ψ across f/H contours is driven by (i) the interaction between baroclinic potential energy and topography (JEBAR), (ii) the curl of wind stress divided by H, (iii) curl of the bottom stress divided by H.

4) Transport Divergence Vorticity Balance

The fourth option is divide the DIM equations (32)–(33) by the Coriolis parameter f and take the curl:

$$\partial_y V + \partial_x U = \partial_y \left[\frac{H}{f} \partial_x p_b + \frac{1}{f} \partial_x \chi - \frac{\mathrm{Ek}}{f} \left(\tau^x + \tau_b^x \right) \right] + \partial_x \left[-\frac{H}{f} \partial_y p_b - \frac{1}{f} \partial_y \chi + \frac{\mathrm{Ek}}{f} \left(\tau^y + \tau_b^y \right) \right], \quad (45)$$

$$\implies 0 = J\left(p_b, \frac{H}{f} \right) + J\left(\chi, \frac{1}{f} \right) + \mathrm{Ek}\, \hat{\mathbf{z}} \cdot \left(\nabla \times \frac{\tau + \tau_b}{f} \right). \quad (46)$$

This equation is called the (linear, time-independent) transport divergence vorticity balance (TDVB; see Waldman and Giordani 2023 Eq. (16) and Nøst and Isachsen 2003 Eq. (13), who call it a form of the "depth-integrated vorticity equation"). The reason is that it is an equation for the divergence of the depth-integrated horizontal flow (which vanishes for time-independent cases). The TDVB does not feature a circulation variable (like U, V, or Ψ). Nevertheless, knowledge of the derivatives of p_b, χ, τ, and τ_b required by the TDVB yields (U, V) from the DIM equations (32)–(33).

c. Remarks on Dynamical Balances

The dynamical balances described in this section (and summarized in Table 1) are useful because they expose relationships between the circulation and the imposed forcing (stress and buoyancy). In some cases, this approach leads to a direct algebraic link. For example, the flat-bottomed BV equations lead to the Sverdrup balance between the depth-integrated advection of planetary vorticity βV and the wind stress curl $\hat{\mathbf{z}} \cdot \nabla \times \tau$ (see (38); vanishing of the curl of the bottom stress $\hat{\mathbf{z}} \cdot \nabla \times \tau_b$ is assumed). As Pedlosky (1996) says, this is "an extraordinary result" that has profound influence on understanding ocean general circulation (see also Pedlosky 1987; Vallis 2006, and Klinger and Haine 2019). Another example concerns the TDVB equation. Upon integration over the region enclosed by an f/H contour, the $J(p_b, f/H)$ term in (46) vanishes. This provides an integral constraint on the relationship between χ, τ, and τ_b. Assuming that the bottom stress τ_b is

simply related to the bottom current, for instance, with a linear drag law, links the circulation to the surface stress and baroclinic potential energy χ. Knowledge of the bottom current then yields the current at other depths via the thermal wind. This balance has been exploited by Nøst and Isachsen (2003); Nilsson et al. (2005, 2024) and Aaboe and Nøst (2008) (see also Timmermans and Marshall 2020).

More generally, the dynamical balances in Table 1 relate circulation variables with unknown fields, like the bottom stress τ_b, the bottom pressure p_b, and the bottom vertical speed $w|_{z=-H}$. These unknown fields are implicitly defined in terms of the forcing τ and b. For this reason, the dynamical balances in Table 1 have been applied *diagnostically* to ocean general circulation model (OGCM) output (for example, see Hughes and Killworth 1995; Borowski et al. 2002; Le Corre et al. 2020; Waldman and Giordani 2023). Ambiguity remains, however, as it is unclear how accurately OGCMs respect the continuous equations they aim to represent. This issue is especially acute in the vicinity of bottom topography that is coarsely resolved (Styles et al. 2022).

The main point of this paper is to show how the original robust diagnostic PG equations can be solved exactly. In this context, solving the equations "exactly" means finding an analytical solution that is coupled to the solution of a two-dimensional elliptic linear partial differential equation for the surface pressure field. In general, that surface pressure field equation is solved numerically. With the solution to the robust diagnostic PG equations in hand, all the diagnostic dynamical balances in Table 1 can be computed exactly, without additional dynamical or geometrical assumptions. This process thereby quantifies the relationship between the three-dimensional velocity and the forcing stress and buoyancy field exactly. It includes the Sverdrup balance and the β-plume as special cases.

4. Solution to Time-Independent Linear Buoyancy Equation

The robust diagnostic approach defined in Sections 1 and 2a requires the relaxation field B and the surface field b_s from which the full buoyancy field b is computed. That computation requires solving the differential equation for b, which is achieved in this section with an analytical formula. The method is prototypical for the method to solve the PG equations in Section 5.

For time-independent, negligible buoyancy advection, and small aspect ratio, the buoyancy equation (15), (17), (19) reads:

$$\mathrm{Ek}\,\partial_z\left(\kappa\partial_z b\right) - \Gamma b = -\Gamma B, \tag{47}$$

$$b_{z=0} = b_s, \tag{48}$$

$$\partial_z b|_{z=-H} = 0, \tag{49}$$

for buoyancy profile $B(x, y, z)$, relaxation coefficient Γ, surface buoyancy field $b_s(x, y)$, and zero vertical buoyancy gradient at the sea bed.[4] Strictly, the surface boundary condition applies at $z = \eta$, but as $|\eta| \ll 1$ (equivalently, the dimensional surface height is very small compared to H_0), it is applied at $z = 0$. The boundary and internal buoyancy sources allow for time-independent solutions that have non-trivial solutions. This problem (47)–(49) is solved exactly by

$$b(z) = -\frac{\Gamma}{\mathrm{Ek}}\int_{-H}^{0} G_b(z;\xi)B(\xi)\,d\xi + \partial_\xi G_b(z;\xi)|_{\xi=0}\, b_s, \tag{50}$$

in which, for uniform vertical diffusivity ($\kappa = 1$),

$$G_b(z;\xi) = \frac{\mathrm{e}^{\psi(z+\xi)}}{2\psi\left(1+\mathrm{e}^{2\psi H}\right)}\begin{cases}\left(1-\mathrm{e}^{-2\psi\xi}\right)\left(\mathrm{e}^{2\psi H}+\mathrm{e}^{-2\psi z}\right) & \text{for } z \le \xi,\\ \left(1-\mathrm{e}^{-2\psi z}\right)\left(\mathrm{e}^{2\psi H}+\mathrm{e}^{-2\psi\xi}\right) & \text{for } z \ge \xi,\end{cases} \tag{51}$$

[4] The more accurate bottom boundary condition on buoyancy, Eq. (19), includes terms like $\partial_x b\,\partial_x H$, which are proportional to the aspect ratio δ squared. They couple the vertical buoyancy variations to the horizontal buoyancy variations. As $\delta \approx 10^{-3} \ll 1$ these effects are small, as assumed in Section 2b. Still, if necessary (for example in the presence of steep topography), they can be handled in what follows using a Steklov-Poincaré operator (Quarteroni and Valli 2023) or a perturbative approach.

with $\psi = \sqrt{\Gamma/\mathrm{Ek}}$. See Appendix A for the derivation of these results and Eq. (A5) for the explicit form of $\partial_\xi G_b(z;\xi)|_{\xi=0}$.

This formula (50) for b is a linear combination of contributions from the relaxation field B and the surface buoyancy b_s. For a particular horizontal position, it depends only on the B profile with depth at that position and the b_s value at that position. This fact remains true for the case where diffusivity κ depends on depth too, but the more general boundary condition (19) introduces non-local coupling between different horizontal positions. One way to view the result (50) is to think of B and b_s as implicitly defining b. The b field results from a sum of B and b_s contributions that are weighted (filtered) by the G_b kernel. This weighting operation ensures that b satisfies the original buoyancy equation and its boundary conditions (which B may not do).

5. Solution to Planetary Geostrophic Equations

This section solves the frictional geostrophic equations by following the approach in Section 4. It relates to textbook Ekman theory (Ekman 1905; Greenspan 1990; Salmon 1998; Vallis 2006), which invokes asymptotic expansions in small Ekman number. The results here are exact, however, and include a water column of finite and variable depth, buoyancy effects, and resolved surface and bottom Ekman layers. Exact results are attainable in the present system because (i) viscous effects only apply in the vertical direction, (ii) resolution of the bottom Ekman layer permits the specification of vanishing flow at the sloping sea floor. These conditions imply that the momentum equations can be solved at each horizontal position independently, given the surface pressure gradient. The surface pressure gradient itself satisfies a two-dimensional differential equation, which is derived at the end of this section.

a. Internal Pressure

Start from the non-dimensional hydrostatic equation (14) to write the pressure p at height z as

$$p = p_s - \int_z^0 b(z')\, dz', \tag{52}$$

$$\implies \partial_x p = \partial_x p_s - \int_z^0 \partial_x b(z')\, dz', \text{ and similarly for } \partial_y p, \tag{53}$$

$$\implies p_b \equiv p|_{z=-H} = p_s - \int_{-H}^0 b(z')\, dz', \tag{54}$$

where p_s is the surface pressure and p_b is the bottom pressure. In (53) the second term on the right hand side due to horizontal variations in the buoyancy field is called the baroclinic pressure gradient.

b. Horizontal Velocity

Writing the horizontal velocity vector as $\mathfrak{u} = u + iv$ (with $i = \sqrt{-1}$), the non-dimensional PG momentum equations (11)–(12), namely, the frictional geostrophic equations, are

$$\mathrm{Ek}\, \partial_z \left(\nu \partial_z \mathfrak{u}\right) - if\mathfrak{u} = \left(\partial_x + i\partial_y\right) p, \tag{55}$$

$$\left[\nu \partial_z \mathfrak{u}\right]_{z=0} = \tau^x + i\tau^y, \tag{56}$$

$$\mathfrak{u}|_{z=-H} = 0. \tag{57}$$

Following the robust diagnostic assumption, the baroclinic part of the pressure gradient is a known forcing function in (55) and the surface pressure gradient is to be determined.

This problem is solved exactly by (see Appendix B for details and `FrictionalGeostrophicEquationSolution.ipynb` #2.1, 2.2)

$$\mathfrak{u}(z) = \frac{1}{\mathrm{Ek}} \int_{-H}^0 G_{\mathfrak{u}}(z;\xi) \left(\partial_x + i\partial_y\right) p(\xi)\, d\xi - G_{\mathfrak{u}}(z;0) \left(\tau^x + i\tau^y\right). \tag{58}$$

For uniform vertical viscosity ($\nu = 1$),

$$G_{\mathfrak{u}}(z;\xi) = \frac{\mathrm{e}^{-\varphi(z+\xi)}}{2\varphi\left(1+\mathrm{e}^{2\varphi H}\right)} \begin{cases} \mathrm{e}^{2\varphi\xi} - \mathrm{e}^{2\varphi(z+H)} - \mathrm{e}^{2\varphi(z+\xi+H)} + 1 & \text{for } z \le \xi, \\ \mathrm{e}^{2\varphi z} - \mathrm{e}^{2\varphi(\xi+H)} - \mathrm{e}^{2\varphi(z+\xi+H)} + 1 & \text{for } z \ge \xi. \end{cases}, \tag{59}$$

$\varphi = \sqrt{i}\,\phi = \sqrt{if/\mathrm{Ek}}$ is a function of y in general because of meridional variations in f, and Equation (B7) gives the formula for $G_{\mathfrak{u}}(z;0)$.

Substituting for the pressure from (53) gives:

$$\begin{aligned} \mathfrak{u} = {} & \frac{1}{\mathrm{Ek}}\left(\partial_x + i\partial_y\right) p_s \int_{-H}^{0} G_{\mathfrak{u}}(z;\xi)\, d\xi \\ & - \frac{1}{\mathrm{Ek}} \int_{-H}^{0} G_{\mathfrak{u}}(z;\xi) \left[\left(\partial_x + i\partial_y\right) \int_{\xi}^{0} b(z')\, dz'\right] d\xi \\ & - G_{\mathfrak{u}}(z;0)\left(\tau^x + i\tau^y\right). \end{aligned} \tag{60}$$

This formula expresses the velocity as being due to the sum of contributions from the surface pressure gradient (first term on the right hand side), the baroclinic pressure gradient (second term), and the surface stress (third term). The integral in the first right hand side term that multiplies the surface pressure gradient is given explicitly by (B6) for the case of uniform viscosity. Recall that $p_s, b, \tau^x + i\tau^y$, and (therefore) $\mathfrak{u}$ are functions of horizontal (x, y) position. The $G_{\mathfrak{u}}$ kernel depends on position through the dependence of f on latitude and also on the variation of depth $H(x, y)$. Apart from the surface pressure, everything in (60) is known: see Section 5e to determine the surface pressure.

c. Bottom Stress

From (60), the bottom stress τ_b can be computed as follows: $\tau_b \equiv -\nu\,\partial_z \mathfrak{u}|_{z=-H}$, where

$$\begin{aligned}\partial_z \mathfrak{u}|_{z=-H} = &\frac{1}{\mathrm{Ek}}\left(\partial_x + i\partial_y\right) p_s \int_{-H}^{0} \partial_z G_{\mathfrak{u}}(z;\xi)|_{z=-H}\, d\xi \\ &- \frac{1}{\mathrm{Ek}} \int_{-H}^{0} \partial_z G_{\mathfrak{u}}(z;\xi)|_{z=-H} \left[\left(\partial_x + i\partial_y\right) \int_{\xi}^{0} b(z')\, dz'\right] d\xi \\ &- \partial_z G_{\mathfrak{u}}(z;0)|_{z=-H}\,(\tau^x + i\tau^y)\,. \end{aligned} \tag{61}$$

The bottom stress sums contributions from the surface pressure gradient, the buoyancy field, and the surface stress. Write this expression using two functions of horizontal position, $S(x,y)$ and $T(x,y)$:

$$\tau_b = \left[\left(\partial_x + i\partial_y\right) p_s\right] S + T. \tag{62}$$

For uniform viscosity, one finds that

$$S = \frac{1}{\varphi\,\mathrm{Ek}} \tanh(\varphi H)\,, \tag{63}$$

$$T = -\frac{1}{\mathrm{Ek}} \int_{-H}^{0} \frac{\mathrm{e}^{\varphi(H-\xi)}}{1+\mathrm{e}^{2\varphi H}} \left(1+\mathrm{e}^{2\varphi\xi}\right) \left[\left(\partial_x + i\partial_y\right) \int_{\xi}^{0} b(z')\, dz'\right] d\xi - \frac{2\mathrm{e}^{\varphi H}}{1+\mathrm{e}^{2\varphi H}} (\tau^x + i\tau^y) \tag{64}$$

(see `FrictionalGeostrophicEquationSolution.ipynb` #2.5 and #2.6). Notice that the direction of the bottom stress depends on position in a non-trivial way: this is discussed further below.

d. Depth-Integrated Flow

The depth-integrated flow $\mathfrak{U}(x,y) \equiv U + iV$ is computed from (25), (52), and (58),

$$\mathfrak{U} = \int_{-H}^{0} \left\{ \frac{1}{\mathrm{Ek}} \int_{-H}^{0} G_{\mathfrak{u}}(z;\xi) \left(\partial_x + i\partial_y\right) \left[p_s - \int_{\xi}^{0} b(z')\, dz'\right] d\xi - G_{\mathfrak{u}}(z;0)\,(\tau^x + i\tau^y) \right\} dz. \tag{65}$$

As for the bottom stress, this formula expresses the depth-integrated flow as a sum of contributions from the surface pressure field, the buoyancy field, and the surface stress. Like the bottom stress

(62), write this expression using two (complex) functions of horizontal position, $\mathfrak{A}(x,y)$ and $\mathfrak{B}(x,y)$:

$$\mathfrak{U} = \left[\left(\partial_x + i\partial_y\right) p_s\right] \mathfrak{A} + \mathfrak{B}. \tag{66}$$

For $\nu = 1$ one finds

$$\mathfrak{A} = \frac{1}{\varphi^2 \,\mathrm{Ek}} \left[\frac{\mathrm{e}^{2\varphi H} - 1}{\varphi\left(1 + \mathrm{e}^{2\varphi H}\right)} - H\right], \tag{67}$$

$$\mathfrak{B} = \frac{1}{\varphi^2\left(1 + \mathrm{e}^{2\varphi H}\right)} \left[\frac{1}{\mathrm{Ek}} \int_{-H}^{0} \mathrm{e}^{-\varphi\xi} \left(\mathrm{e}^{\varphi H} - \mathrm{e}^{\varphi\xi}\right)\left(\mathrm{e}^{\varphi(\xi+H)} - 1\right)\left[\left(\partial_x + i\partial_y\right) \int_{\xi}^{0} b(z')\, dz'\right] d\xi \right.$$
$$\left. + \left(\mathrm{e}^{\varphi H} - 1\right)^2 \left(\tau^x + i\tau^y\right)\right] \tag{68}$$

(see `FrictionalGeostrophicEquationSolution.ipynb` #2.5 and #2.7). Note also that the streamfunction for the depth-integrated flow (44) is given by:

$$\begin{aligned} \nabla^2 \Psi &= -\left(i\partial_x + \partial_y\right) \mathfrak{U}, \\ &= -\left(i\partial_x + \partial_y\right)\left[\mathfrak{A}\left(\partial_x + i\partial_y\right) p_s + \mathfrak{B}\right] \end{aligned} \tag{69}$$

with Ψ a constant on domain boundaries where $H = 0$.

e. Surface Pressure

In the expressions for the horizontal velocity (60), the bottom stress (62), the depth-integrated flow (66), and its streamfunction (69), the surface pressure gradient contributes, but is not yet specified. How is the surface pressure found? To find the pressure and close the problem, utilize the constraint (13) that the vertically-integrated flow is non-divergent (Salmon 1998). This requirement amounts to (from (37)):

$$\Re\left\{\left(\partial_x - i\partial_y\right) \mathfrak{U}\right\} = 0 \tag{70}$$

($\mathfrak{R}$ is the real part), which implies that

$$\mathfrak{R}\left\{ \left(\partial_x - i\partial_y\right) \left\{ \left[\left(\partial_x + i\partial_y\right) p_s \right] \mathfrak{A} + \mathfrak{B} \right\} \right\} = 0 \tag{71}$$

from (66). The equation for the surface pressure is thus

$$\mathfrak{R}\left\{ \partial_x \left[\mathfrak{A}\, \partial_x p_s + i\mathfrak{A}\, \partial_y p_s \right] - i\partial_y \left[\mathfrak{A}\, \partial_x p_s + i\mathfrak{A}\, \partial_y p_s \right] \right\} = -\mathfrak{R}\left\{ \left(\partial_x - i\partial_y\right) \mathfrak{B} \right\}. \tag{72}$$

In more compact form, this is the real parts of:

$$\left[\mathfrak{A}\nabla^2 + \partial_x \mathfrak{A}\, \partial_x + \partial_y \mathfrak{A}\, \partial_y + i \left(\partial_x \mathfrak{A}\, \partial_y - \partial_y \mathfrak{A}\, \partial_x \right) \right] p_s = - \left(\partial_x - i\partial_y\right) \mathfrak{B}. \tag{73}$$

This is a second order elliptic partial differential equation for p_s given the surface stress and the buoyancy field (which are in $\mathfrak{B}$ from (68)). In general (for arbitrary depth H), this problem must be solved numerically (see Appendix C and examples below).

Still, insight is obtained by noticing that (73) can be written as

$$\nabla \left(\mathfrak{A}^r \nabla p_s \right) - J\left(\mathfrak{A}^i, p_s \right) = -\partial_x \mathfrak{B}^r - \partial_y \mathfrak{B}^i, \tag{74}$$

where the real and imaginary parts of functions $\mathfrak{A}$ and $\mathfrak{B}$ are $\mathfrak{A} = \mathfrak{A}^r + i\mathfrak{A}^i$, $\mathfrak{B} = \mathfrak{B}^r + i\mathfrak{B}^i$, and Jacobian $J(\mathfrak{A}^i, p_s) = \partial_x \mathfrak{A}^i\, \partial_y p_s - \partial_y \mathfrak{A}^i\, \partial_x p_s$. This is a forced advection-diffusion equation for the surface pressure: p_s is advected by a velocity $(\partial_y \mathfrak{A}^i, -\partial_x \mathfrak{A}^i)$, diffused with diffusivity $\mathfrak{A}^r$, and forced with the convergence of the $(\mathfrak{B}^r, \mathfrak{B}^i)$ field. Eq. (67) shows that one part of $\mathfrak{A}^i$ is H/f (recall that $\varphi^2 = if/\mathrm{Ek}$), indicating advection of p_s along f/H contours, as expected. Similarly, the equation for the streamfunction of the depth-integrated flow is a forced advection-diffusion equation. This extends the GCVB (eq. (43)) because it has explicitly included the bottom stress by solving the frictional geostrophic equation.

6. Results

CONTINUE HERE!

Examples of buoyancy solutions b and buoyancy Green's functions G_b are plotted in Figure 1. Two Ekman numbers are shown (Ek = 0.25 and 0.025), according to the row of the subplot, and $H = \Gamma = \kappa = 1$ (so $\psi = \mathrm{Ek}^{-1/2}$ here). Four example b profiles are shown for combinations of $b_s = 0$ and $b_s = 0.1$, and $B(z) = z$ and $B(z) = \mathrm{e}^{4z/H} - 1$ (the $B(z)$ relaxation profiles are also plotted in Figure 1a, c). The b solutions match the b_s surface value and the no flux bottom boundary condition (49). They are drawn towards the $B(z)$ profiles, especially for small Ek, as expected from the buoyancy equation (47). The b profiles smoothly interpolate between these constraints. Figure 1b, d shows the $G_b(z;\xi)$ Green's functions for various ξ source depths. They are peaked (discontinuous) at $z = \xi$ with larger magnitudes and narrower peaks for smaller Ekman number. These functions weight the contribution from the relaxation profile B in (50). Figure 1 also shows the $\partial_\xi G_b(z;\xi)|_{\xi=0}$ functions from (A5). They equal one at the surface and decay with depth, more rapidly for smaller Ekman number. These functions multiply the surface boundary condition b_s in (50).

The horizontal velocity Green's function G_{u} is plotted in Figure 2. Two Ekman numbers are shown (Ek = 0.25 and 0.025), according to the row of the subplot, and $H = f = \nu = 1$ (so $\phi = \mathrm{Ek}^{-1/2}$ here). The G_{u} magnitude is shown in the left column, and the phase in the right column. Figure 2 also shows the vertically-integrated G_{u} from (B6). To interpret these results, first consider the case of vanishing pressure gradient and surface stress in the positive x direction (eastward; $\tau^y = 0$). Eq. (60) shows that the surface velocity is then $-G_{\mathrm{u}}(0;0)\tau^x$. The component $G_{\mathrm{u}}(0;0)$ is on the $\xi/H = 0$ curve, and has a phase of about 135° for Ek = 0.025 (Figure 2d). Therefore, $-G_{\mathrm{u}}(0;0)$ has a phase of -45°, which is measured anti-clockwise from the direction of the stress (see `FrictionalGeostrophicEquationSolution.ipynb`). This corresponds to a surface current at 45° to the right of the stress (in the northern hemisphere), consistent with small Ekman number theory (Salmon 1998; Vallis 2006). Next consider the case of vanishing surface

stress and an imposed surface pressure gradient in the positive x direction (sea level slopes up to the east; $p_s = p_s^x$ for real, positive p_s^x). From (60) the surface velocity is then proportional to $\int_{-H}^{0} G_{\mathrm{u}}(z;\xi)\, d\xi$. The component $\int_{-H}^{0} G_{\mathrm{u}}(z;\xi)\, d\xi$ is labeled Gu_int and has a phase of about 90° for Ek = 0.025 (Figure 2d). This corresponds to a surface geostrophic current flowing northwards with the pressure gradient force pushing westwards (opposite to the pressure gradient itself) and an equal and opposite Coriolis force pushing eastwards. Finally, for the same case (no stress, $p_s = p_s^x$), consider the bottom stress. From (62) and (63), τ_b is proportional to $(1/\varphi)\tanh(\varphi H)$. Thus, as Ek gets small $\varphi = \sqrt{if/\mathrm{Ek}}$ diverges, $\tanh(\varphi H) \to 1$ and τ_b becomes proportional to $i^{-1/2} = 1 - i$. The bottom stress $1 - i$ corresponds to a southeast direction, which is at 135° to the direction of the geostrophic current, consistent with textbook theory (Vallis 2006).

These results establish the validity of the horizontal velocity Green's function G_{u} in Figure 2 for familiar, special cases of (60). More generally, Figure 2 shows that G_{u} respects the homogeneous velocity boundary conditions (56) and (57) of vanishing vertical derivative at the surface and vanishing magnitude at the bottom (see (B2) and (B3)). The (discontinuous) peaks at $z = \xi$ in $G_{\mathrm{u}}(z;\xi)$ show how a pressure gradient causes a response in the horizontal velocity that is at 135° to the pressure gradient; that is, at -45° to the pressure gradient force (in the northern hemisphere). Therefore, the water moves at 45° to the right of the pressure gradient force. This response decays with height above and below $z = \xi$ in clockwise spirals that represent internal Ekman layers. The decay length scale depends inversely on the Ekman number, as expected. The influence of the surface pressure gradient extends throughout the water column until it decays to zero in the bottom Ekman layer (Figure 2 Gu_int curves).

Useful for comparison: Willmott and Luneva (2015)

For an unstratified, flat-bottomed fluid ($\chi = 0, \nabla p_b = \nabla p_s, H = H_0$), specifying the surface pressure gradient ∇p_s closes the problem. We compute the surface stress from (**??**) by substituting

(**??**). Then $u(z), v(z), U, V$ and the force balance can all be computed. See Figure 3 for an example calculation. Alternatively, the surface stress can be specified, and the pressure gradient can be computed (see `Example_barotropic_GeostrophicFlow_v0.3.ipynb`). In practice, it seems that the bottom stress always vanishes. Why?? Is this correct??

Next steps are to solve for:

1. $b = 0, H(x,y) = 1 - x^2 - y^2$ in a circular parabolic basin. See `ExampleSolution_v0.3.ipynb`. DONE

2. $b(x,y)$ given by solving the vertical diffusive buoyancy equation (see Section 4) and $H(x,y) = 1 - x^2 - y^2$ in a circular parabolic basin. DONE. For `SymPy` version, see `ExampleSolution_0.5.ipynb`, for `Symbolics` version, see `ExampleSolution_0.6.ipynb`. Note that the talk for the November 2025 Arctic Dynamics Workshop in Paris used `ExampleSolution_0.4.ipynb`, which has an error in the computation of $\mathfrak{u}(z)$.

3. Add computation of Ψ. See `ExampleSolution_0.7.ipynb`. DONE.

4. Work example with non-trivial bathymetry, like a basin separated by ridge. See `ExampleSolution_0.8.ipynb`. DONE.

5. Add variable f. Work mid-latitude β-plane example with flat bottom. See `ExampleSolution_0.9.ipynb`. DONE.

6. Re-dimensionalize.

7. Discussion

- Talk about including horizontal friction too.

- Topostrophy (Merryfield and Scott 2007; Holloway 2008)
- This diagnostic quantification could be used, for instance, as a baseline to compare with solutions that transcend PG dynamics, like those that include momentum advection, time-dependence, and/or horizontal dissipation.
- Spin up of OGCMs.
- Faster external mode solver.

Acknowledgments. Large language models (GPT 5.4 mini, GPT 5.5, Github Copliot version 1.1.21) were used for assistance with Julia code, especially debugging and plotting of final figures, and some literature exploration and code generation. The analytical results were supported by the `SymPy` and `Symbolics.jl` computer algebra systems (Meurer et al. 2017; Gowda et al. 2021).

Data availability statement. All codes to reproduce the analytical and numerical results in this paper are available at `github.com/ThomasHaine/PlanetaryGeostrophyAnalyticalModel`.

8. Appendices

APPENDIX A

Solution to the linear time-independent buoyancy equation

The problem (47)–(49) is solved as follows using a Green's function G_b. The Green's function satisfies:

$$\partial_z\left(\kappa\partial_z G_b\right) - \frac{\Gamma}{\mathrm{Ek}} G_b = \delta(z-\xi), \tag{A1}$$

$$\text{Surface } z = 0 : G_b = 0, \tag{A2}$$

$$\text{Bottom } z = -H : \partial_z G_b = 0, \tag{A3}$$

where $\delta(z-\xi)$ is the Dirac delta function. This is a second order linear ordinary differential equation with two boundary conditions, which can be solved exactly, at least for certain diffusivity profiles. The solution for uniform non-dimensional diffusivity is (see `BuoyancyEquationSolution.ipynb` #1.1, #1.11)

$$G_b(z;\xi) = \frac{\mathrm{e}^{\psi(z+\xi)}}{2\kappa\psi\left(1+\mathrm{e}^{2\psi H}\right)} \begin{cases} \left(1-\mathrm{e}^{-2\psi\xi}\right)\left(\mathrm{e}^{2\psi H}+\mathrm{e}^{-2\psi z}\right) & \text{for } z \le \xi, \\ \left(1-\mathrm{e}^{-2\psi z}\right)\left(\mathrm{e}^{2\psi H}+\mathrm{e}^{-2\psi\xi}\right) & \text{for } z \ge \xi, \end{cases} \tag{A4}$$

with $\psi = \sqrt{\Gamma/(\kappa\,\mathrm{Ek})}$. To compute the contribution from the surface buoyancy field b_s to b we need the ξ derivative of the buoyancy Green's function at the surface, which is

$$\partial_\xi G_b(z;\xi)|_{\xi=0} = \frac{\mathrm{e}^{\psi z}\left(\mathrm{e}^{2\psi H}+\mathrm{e}^{-2\psi z}\right)}{\kappa\left(1+\mathrm{e}^{2\psi H}\right)}. \tag{A5}$$

(see `BuoyancyEquationSolution.ipynb` #1.2). These results lead to (50). If required, the solution for a bottom-intensified exponential diffusivity profile, like Eq. (17) of Peterson and Callies (2026), can be found using hypergeometric functions (Arfken and Weber 1995). The form of the solution (50) is the same, just the Green's function G_b differs.

In solving the momentum equations, one needs horizontal derivatives of the vertically-integrated buoyancy field to compute the pressure gradient (53). We need these two terms:

$$\begin{aligned} \int_\xi^0 G_b(z';\xi')\,dz' = \frac{1}{2\kappa\psi^2\left(1+\mathrm{e}^{2\psi H}\right)}\Big[\\ \mathrm{e}^{\psi(\xi-\xi'+2H)} - \mathrm{e}^{\psi(\xi+\xi'+2H)} + 2\mathrm{e}^{\psi(\xi'+2H)} + 2\mathrm{e}^{-\xi'\psi} + \mathrm{e}^{\psi(\xi'-\xi)} - \mathrm{e}^{-\psi(\xi+\xi')} - \\ \mathrm{e}^{\psi(\xi'+2H-\mathrm{Max}(\xi,\xi'))} - \mathrm{e}^{-\psi(\xi'-2H-\mathrm{Max}(\xi,\xi'))} - \mathrm{e}^{\psi(\xi'-\mathrm{Max}(\xi,\xi'))} - \mathrm{e}^{-\psi(\xi'-\mathrm{Max}(\xi,\xi'))}\Big] \equiv G_{b,1}(\xi,\xi') \end{aligned} \tag{A6}$$

and

$$\int_\xi^0 \partial_\xi G_b(z';\xi)|_{\xi=0}\,dz' = \frac{\mathrm{e}^{2\psi H}+\mathrm{e}^{-\xi\psi}-\mathrm{e}^{\psi(\xi+2H)}-1}{\kappa\psi\left(1+\mathrm{e}^{2\psi H}\right)} \equiv G_{b,2}(\xi) \tag{A7}$$

(see `BuoyancyEquationSolution.ipynb`, #1.3, #1.31).

APPENDIX B

Solution to the frictional geostrophic equations

Velocity Green's function

The Green's function corresponding to (55)–(57) satisfies:

$$\partial_z\left(\nu\partial_z G_{\mathfrak{u}}\right) - \frac{if}{\mathrm{Ek}} G_{\mathfrak{u}} = \delta(z-\xi), \tag{B1}$$

$$\left(\nu\partial_z G_{\mathfrak{u}}\right)\big|_{z=0} = 0, \tag{B2}$$

$$G_{\mathfrak{u}}\big|_{z=-H} = 0. \tag{B3}$$

We find that the solution for uniform non-dimensional viscosity ν is:

$$G_{\mathfrak{u}}(z;\xi) = \frac{\mathrm{e}^{-\varphi(z+\xi)}}{2\nu\,\varphi\left(1+\mathrm{e}^{2\varphi H}\right)} \begin{cases} \mathrm{e}^{2\varphi\xi} - \mathrm{e}^{2\varphi(z+H)} - \mathrm{e}^{2\varphi(z+\xi+H)} + 1 & \text{for } z \le \xi, \\ \mathrm{e}^{2\varphi z} - \mathrm{e}^{2\varphi(\xi+H)} - \mathrm{e}^{2\varphi(z+\xi+H)} + 1 & \text{for } z \ge \xi \end{cases} \tag{B4}$$

which is (59) for $\nu = 1$. In this formula $\varphi = \sqrt{i}\phi = \sqrt{if/(\nu\,\mathrm{Ek})}$ (see `FrictionalGeostrophicEquationSolution.ipynb` #1.1). As for the buoyancy equation in Appendix A, the Green's function for exponential viscosity profiles can be found using hypergeometric functions.

Solution for velocity field

From (60), the velocity field is:

$$\begin{aligned} \mathfrak{u} = {} & \frac{1}{\mathrm{Ek}}\left(\partial_x + i\partial_y\right) p_s \int_{-H}^{0} G_{\mathfrak{u}}(z;\xi)\, d\xi - G_{\mathfrak{u}}(z;0)\left(\tau^x + i\tau^y\right) \\ & - \frac{1}{\mathrm{Ek}} \int_{-H}^{0} G_{\mathfrak{u}}(z;\xi) \Bigg[\left(\partial_x + i\partial_y\right) \underbrace{\int_{\xi}^{0} b(z')\, dz'}_{\Pi(\xi)} \Bigg] d\xi. \end{aligned} \tag{B5}$$

In what follows, remember that p_s, τ^x, τ^y, b (from B and b_s), H (and hence $G_{\mathfrak{u}}$ and G_b) are all functions of horizontal position (x, y). This explicit dependence is suppressed for notational clarity.

The first term on the right hand side of (B5) is a product of the surface pressure gradient and the known function

$$\int_{-H}^{0} G_{\mathfrak{u}}(z;\xi)\, d\xi = \frac{\mathrm{e}^{\varphi(z+H)} + \mathrm{e}^{\varphi(H-z)} - \mathrm{e}^{2\varphi H} - 1}{\nu\,\varphi^2\left(1+\mathrm{e}^{2\varphi H}\right)} \equiv G_{\mathfrak{u},1}(z) \tag{B6}$$

(see `FrictionalGeostrophicEquationSolution.ipynb` #1.2). The second term is a product of the surface stress and another known function:

$$G_{\mathfrak{u}}(z;0) = \frac{\mathrm{e}^{-\varphi z}}{\nu\,\varphi\left(1+\mathrm{e}^{2\varphi H}\right)}\left[1-\mathrm{e}^{2\varphi(z+H)}\right] \equiv G_{\mathfrak{u},2}(z) \tag{B7}$$

(see `FrictionalGeostrophicEquationSolution.ipynb` #1.3). The last term, which is the baroclinic pressure gradient, is more complicated. We rewrite it in a more convenient form, as follows. Substitute using (50)

$$\begin{aligned}
\Pi(\xi) &\equiv \int_{\xi}^{0} b(z')\, dz', \\
&= \int_{\xi}^{0}\left[-\frac{\Gamma}{\mathrm{Ek}}\int_{-H}^{0} G_b(z';\xi')B(\xi')\, d\xi' + \partial_\xi G_b(z';\xi)|_{\xi=0}\, b_s\right] dz', \\
&= -\frac{\Gamma}{\mathrm{Ek}}\int_{-H}^{0}\int_{\xi}^{0} G_b(z';\xi')\, dz'\, B(\xi')\, d\xi' + \int_{\xi}^{0} \partial_\xi G_b(z';\xi)|_{\xi=0}\; dz'\, b_s, \qquad \text{(B8)} \\
&= \int_{-H}^{0} G_{b,1}(\xi,\xi')B(\xi')\, d\xi' + G_{b,2}(\xi)\, b_s, \qquad \text{(B9)}
\end{aligned}$$

where $G_{b,1}$ and $G_{b,2}$ are given by (A6) and (A7). Hence, (B5) becomes:

$$\begin{aligned}
\mathrm{Ek}\,\mathfrak{u} = {} & G_{\mathfrak{u},1}(z)\left(\partial_x + i\partial_y\right) p_s - \mathrm{Ek}\, G_{\mathfrak{u},2}(z)\left(\tau^x + i\tau^y\right) \\
& - \int_{-H}^{0} G_{\mathfrak{u}}(z;\xi)\left[\left(\partial_x + i\partial_y\right)\left[\int_{-H}^{0} G_{b,1}(\xi,\xi')B(\xi')\, d\xi' + G_{b,2}(\xi)\, b_s\right]\right] d\xi. \qquad \text{(B10)}
\end{aligned}$$

This formula can be used to compute $\mathfrak{u}$, but it involves a double integral in the vertical direction (over ξ and ξ') for every (x, y, z) point. An alternative formula is derived as follows.

Consider the terms in (B5) involving the x and y derivatives of Π. The y derivative term is

$$\begin{aligned}\int_{-H}^{0} G_{\mathfrak{u}}(z;\xi)\,\partial_y\Pi(y,\xi)\,d\xi &= \int_{-H}^{0}\partial_y\left[G_{\mathfrak{u}}(z;\xi)\,\Pi(y,\xi)\right]\,d\xi - \int_{-H}^{0}\left[\partial_y G_{\mathfrak{u}}(z;\xi)\right]\Pi(y,\xi)\,d\xi \\ &= \partial_y\left[\int_{-H}^{0} G_{\mathfrak{u}}(z;\xi)\,\Pi(y,\xi)\,d\xi\right] - \left[\partial_y H\right]G_{\mathfrak{u}}(z;-H)\,\Pi(y,-H) \\ &\quad - \int_{-H}^{0}\left[\partial_y G_{\mathfrak{u}}(z;\xi)\right]\Pi(y,\xi)\,d\xi \end{aligned} \tag{B11}$$

using Leibniz' integration formula (footnote 1) and recalling that $G_{\mathfrak{u}}$ depends on position y via the non-dimensional Coriolis parameter in φ and on depth $H(x,y)$.

The middle term on the right hand side of (B11) is straightforward as

$$G_{\mathfrak{u}}(z;-H) = -\frac{\sinh\varphi\,(H+z)}{\nu\varphi}. \tag{B12}$$

The first term on the right hand side of (B11) is a sum of two terms,

$$\begin{aligned}&\int_{-H}^{0} G_{\mathfrak{u}}(z;\xi)\,\Pi(x,y,\xi)\,d\xi = \\ &\int_{-H}^{0} G_{\mathfrak{u}}(z;\xi)\int_{\xi}^{0}\left[-\frac{\Gamma}{\mathrm{Ek}}\int_{-H}^{0} G_b(z';\xi')B(\xi')\,d\xi' + \partial_\xi G_b(z';\xi)|_{\xi=0}\,b_s\right]dz'\,d\xi. \end{aligned} \tag{B13}$$

The order of integration can be switched so that the triple integral involving relaxation to B becomes

$$\begin{aligned}&-\frac{\Gamma}{\mathrm{Ek}}\int_{-H}^{0} G_{\mathfrak{u}}(z;\xi)\int_{\xi}^{0}\int_{-H}^{0} G_b(z';\xi')B(x,y,\xi')\,d\xi'\,dz'\,d\xi = \\ &-\frac{\Gamma}{\mathrm{Ek}}\int_{-H}^{0}\left[\int_{-H}^{0} G_{\mathfrak{u}}(z;\xi)\int_{\xi}^{0} G_b(z';\xi')\,dz'\,d\xi\right]B(\xi')\,d\xi' \equiv \int_{-H}^{0} G_{\mathfrak{u},b}(z;\xi')B(\xi')\,d\xi', \end{aligned} \tag{B14}$$

where

$$G_{\mathfrak{u},b}(z;\xi') = -\frac{\Gamma}{\mathrm{Ek}}\int_{-H}^{0} G_{\mathfrak{u}}(z;\xi)\int_{\xi}^{0} G_b(z';\xi')\,dz'\,d\xi. \tag{B15}$$

The formula for $G_{\mathfrak{u},b}$ is computed using `SymPy.jl`, but it is too long to write out here. It is converted to a `Symbolics.jl` function in `ComputeQuadratures.ipynb` and is tested against numerical quadrature in `SymbolicsTesting.jl`. The `Symbolics.jl` $G_{\mathfrak{u},b}$ function is around

10^3 times faster than the numerical approximation, and they agree to many significant figures over a wide range of Ek and ξ' values. The double integral b_s term in (B13) is

$$\int_{-H}^{0} G_{\mathrm{u}}(z;\xi) \int_{\xi}^{0} \partial_\xi G_b(z';\xi)|_{\xi=0}\, b_s \, dz' \, d\xi \equiv G_{\mathrm{u},b_\xi}(z)\, b_s, \tag{B16}$$

where

$$G_{\mathrm{u},b_\xi}(z) = \int_{-H}^{0} G_{\mathrm{u}}(z;\xi) \int_{\xi}^{0} \partial_\xi G_b(z';\xi)|_{\xi=0} \, dz' \, d\xi. \tag{B17}$$

As for $G_{\mathrm{u},b}$, the expression for G_{u,b_ξ} is computed using `SymPy.jl` (see `ComputeQuadratures.ipynb`) and tested in `SymbolicsTesting.jl`. Hence, the first term on the right hand side of (B11) is

$$\int_{-H}^{0} G_{\mathrm{u}}(z;\xi)\,\Pi(\xi)\, d\xi = \int_{-H}^{0} G_{\mathrm{u},b}(z;\xi')B(\xi')\, d\xi' + G_{\mathrm{u},b_\xi}(z)\, b_s. \tag{B18}$$

Finally, we need the third term on the right hand side of (B11):

$$\begin{aligned}\int_{-H}^{0} \left[\partial_y G_{\mathrm{u}}(z;\xi)\right] \Pi(x,y,\xi)\, d\xi = \int_{-H}^{0} \partial_y G_{\mathrm{u}}(z;\xi) \int_{\xi}^{0} \Bigg[-\frac{\Gamma}{\mathrm{Ek}} \int_{-H}^{0} G_b(z';\xi')B(x,y,\xi')\, d\xi' \\ + \partial_\xi G_b(z';\xi)|_{\xi=0}\, b_s(x,y) \Bigg] \, dz' \, d\xi \end{aligned} \tag{B19}$$

Like (B13) this consists of two pieces:

$$\int_{-H}^{0} \left[\partial_y G_{\mathrm{u}}(z;\xi)\right] \Pi(x,y,\xi)\, d\xi = \int_{-H}^{0} G_{\mathrm{u}_y,b}(z;\xi')B(x,y,\xi')\, d\xi' + G_{\mathrm{u}_y,b_\xi}(z)\, b_s(x,y), \tag{B20}$$

where

$$G_{\mathrm{u}_y,b}(z;\xi') = -\frac{\Gamma}{\mathrm{Ek}} \int_{-H}^{0} \partial_y G_{\mathrm{u}}(z;\xi) \int_{\xi}^{0} G_b(z';\xi') \, dz' \, d\xi, \tag{B21}$$

$$G_{\mathrm{u}_y,b_\xi}(z) = \int_{-H}^{0} \partial_y G_{\mathrm{u}}(z;\xi) \int_{\xi}^{0} \partial_\xi G_b(z';\xi)|_{\xi=0} \, dz' \, d\xi. \tag{B22}$$

These terms are also computed using `SymPy.jl` (see `ComputeQuadratures.ipynb`) and tested in `SymbolicsTesting.jl`. The corresponding x derivative kernels, $G_{\mathrm{u}_x,b}(z;\xi')$ and $G_{\mathrm{u}_x,b_\xi}(z)$ are defined analogously.

Thus, (B5) leads to

$$
\begin{aligned}
\mathrm{Ek}\,\mathfrak{u} = {} & \left[\left(\partial_x + i\partial_y\right) p_s\right] \underbrace{G_{\mathfrak{u},1}(z)}_{\text{(B6)}} - \mathrm{Ek}\, \underbrace{G_{\mathfrak{u},2}(z)}_{\text{(B7)}} \left(\tau^x + i\tau^y\right) \\
& - \left(\partial_x + i\partial_y\right) \left[\int_{-H}^{0} \underbrace{G_{\mathfrak{u},b}(z;\xi')}_{\text{(B15)}} B(\xi')\, d\xi' + \underbrace{G_{\mathfrak{u},b_\xi}(z)}_{\text{(B17)}}\, b_s \right] \\
& + \left[\left(\partial_x + i\partial_y\right) H\right] \underbrace{G_{\mathfrak{u}}(z;-H)}_{\text{(B12)}} \left[\int_{-H}^{0} \underbrace{G_{b,1}(-H,\xi')}_{\text{(A6)}} B(\xi')\, d\xi' + \underbrace{G_{b,2}(-H)}_{\text{(A7)}}\, b_s \right] \\
& + \int_{-H}^{0} \underbrace{G_{\mathfrak{u}_x,b}(z;\xi')}_{\text{from (B21)}} B(\xi')\, d\xi' + \underbrace{G_{\mathfrak{u}_x,b_\xi}(z)}_{\text{from (B22)}}\, b_s \\
& + i \int_{-H}^{0} \underbrace{G_{\mathfrak{u}_y,b}(z;\xi')}_{\text{(B21)}} B(\xi')\, d\xi' + i \underbrace{G_{\mathfrak{u}_y,b_\xi}(z)}_{\text{(B22)}}\, b_s .
\end{aligned}
\tag{B23}
$$

Compared to (B10), this formula involves (four) single integrals in the vertical direction for every (x, y, z) point. Although the latter expression for velocity is more complicated than the former one, it is 25 times faster to compute for typical parameters. Moreover, the two versions are compared to each other to test the derivation and code: they agree to many significant figures (see `RunVelocityFieldPrototype.jl`).

APPENDIX C

Numerical methods

a. Numerical solution to the surface pressure and depth-integrated streamfunction

To compute solutions for the surface pressure field p_s numerically, the weak form of (73) is required, as follows. Take the divergence form of the equation, multiply by test function v, and

integrate over all space:

$$\int_{x,y} v\left(\partial_x - i\partial_y\right)\left[\mathfrak{A}\left(\partial_x + i\partial_y\right)p_s\right]\, dy\, dx = \int_{x,y} v\left(-\partial_x + i\partial_y\right)\mathfrak{B}\, dy\, dx,$$
$$\implies \int_{x,y} \left(\partial_x - i\partial_y\right) v\left[\mathfrak{A}\left(\partial_x + i\partial_y\right)p_s\right]\, dy\, dx = \int_{x,y} \mathfrak{B}\left(-\partial_x + i\partial_y\right) v\, dy\, dx \tag{C1}$$

assuming that p_s and v vanish on the boundaries.

Similarly, the weak form to find the streamfunction for the depth-integrated flow Ψ is (from (69)):

$$\int_{x,y} v\nabla^2\Psi\, dy\, dx = \int_{x,y} v\left(-\left(i\partial_x + \partial_y\right)\left[\mathfrak{A}\left(\partial_x + i\partial_y\right)p_s + \mathfrak{B}\right]\right)\, dy\, dx,$$
$$\implies \int_{x,y} \nabla\Psi\cdot\nabla v\, dy\, dx = -\int_{\Omega} \mathfrak{C}\left(i\partial_x + \partial_y\right)v\, dy\, dx,$$

where

$$\mathfrak{C} = \mathfrak{A}(\partial_x + i\partial_y)p_s + \mathfrak{B}, \tag{C2}$$

assuming that Ψ and v vanish on the boundaries. These weak form equations are used to construct solutions using the `gridap.jl` Julia package (Badia and Verdugo 2020).

b. Solution validation

In order to test and validate the derivation of the solution method, the results for a test case are compared to an independently-derived solution. We use the `Oceananigans.jl` Julia package, which solves the Boussinesq incompressible nonhydrostatic Navier-Stokes equations (Wagner et al. 2025). The test involves an elliptical basin with smoothly-varying topography on a β-plane. The forcing consists of a zonal windstress that varies in the meridional direction, and varying surface buoyancy and relaxation buoyancy fields. The forcing amplitude is small so that buoyancy and momentum advection are weak and the linear PG equations are an accurate asymptotic approximation.

Solutions for the surface pressure, buoyancy, and velocity fields from the Oceananigans solution and the PG solution are compared (see `Oceananigans_Variable_Basin_Validation.ipynb`). The differences between the two solutions are of order 1% and diminish as the grid spacing in the Oceananigans model is refined. On this basis, we presume that the PG solution is derived and coded accurately. The test is important as parts of the PG solution are produced automatically using `Sympy.jl` and `Symbolics.jl`, and some of the software development employs artificial intelligence assistance (Github Copilot version 1.1.21). For typical parameters, computing the PG solution is hundreds of times faster than computing the Oceananigans solution.

References

Aaboe, S., and O. A. Nøst, 2008: A diagnostic model of the Nordic Seas and Arctic Ocean circulation: Quantifying the effects of a variable bottom density along a sloping topography. *J. Phys. Oceanogr.*, **38 (12)**, 2685–2703, doi:10.1175/2008jpo3862.1.

Anderson, D. L., and A. Gill, 1975: Spin-up of a stratified ocean with applications to upwelling. *Deep Sea Res.*, **22 (9)**, 583–596, doi:10.1016/0011-7471(75)90046-7.

Arfken, G. B., and H. J. Weber, 1995: *Mathematical methods for physicists*. 4th ed., Academic Press, San Diego, San Diego.

Badia, S., and F. Verdugo, 2020: Gridap: An extensible finite element toolbox in Julia. *J. Open Source Soft.*, **5 (52)**, 2520, doi:10.21105/joss.02520.

Bogden, P. S., R. E. Davis, and R. Salmon, 1993: The North Atlantic circulation: Combining simplified dynamics with hydrographic data. *J. Mar. Res.*, **51 (1)**, 1–52, doi:10.1357/0022240933223855.

Borowski, D., R. Gerdes, and D. Olbers, 2002: Thermohaline and wind forcing of a circumpolar channel with blocked geostrophic contours. *J. Phys. Oceanogr.*, **32 (9)**, 2520–2540, doi:10.1175/1520-0485(2002)032⟨2520:tawfoa⟩2.0.co;2.

de Verdière, C. A., 1988: Buoyancy driven planetary flows. *J. Mar. Res.*, **46 (2)**, 215–265, doi:10.1357/002224088785113667, URL https://elischolar.library.yale.edu/journal_of_marine_research/1887.

Ekman, V. W., 1905: On the influence of the earth's rotation on ocean-currents. *Arkiv for Matematik, Astronomi och Fysik*, **2**, 1–52.

Gowda, S., Y. Ma, A. Cheli, M. Gwóźzdź, V. B. Shah, A. Edelman, and C. Rackauckas, 2021: High-performance symbolic-numerics via multiple dispatch. *ACM Communications in Computer Algebra*, **55 (3)**, 92–96, doi:10.1145/3511528.3511535.

Greenspan, H. P., 1990: *The theory of rotating fluids*. Breuklen Press, Brookline, MA, USA, First published in 1968 by Cambridge University Press as part of the Cambridge Monographs on Mechanics and Applied Mathematics.

Holloway, G., 2008: Observing global ocean topostrophy. *J. Geophys. Res.*, **113 (C7)**, doi:10.1029/2007jc004635.

Hughes, C. W., and P. D. Killworth, 1995: Effects of bottom topography in the large-scale circulation of the Southern Ocean. *J. Phys. Oceanogr.*, **25 (11)**, 2485–2497, doi:10.1175/1520-0485(1995)025⟨2485:eobtit⟩2.0.co;2.

Klinger, B. A., and T. W. N. Haine, 2019: *Ocean Circulation in Three Dimensions*. 1st ed., Cambridge University Press, Cambridge, United Kingdom and New York, NY, USA, URL http://www.cambridge.org/9780521768436.

584 Le Corre, M., J. Gula, and A.-M. Tréguier, 2020: Barotropic vorticity balance of the North
585 Atlantic subpolar gyre in an eddy-resolving model. *Ocean Sci.*, **16 (2)**, 451–468, doi:10.5194/
586 os-16-451-2020.

587 Marshall, D., 1995: Influence of topography on the large-scale ocean circulation.
588 *J. Phys. Oceanogr.*, **25 (7)**, 1622–1635, doi:10.1175/1520-0485(1995)025⟨1622:iototl⟩2.0.co;2.

589 Merryfield, W. J., and R. B. Scott, 2007: Bathymetric influence on mean currents in two high-
590 resolution near-global ocean models. *Ocean Modelling*, **16 (1-2)**, 76–94, doi:10.1016/j.ocemod.
591 2006.07.005.

592 Meurer, A., and Coauthors, 2017: Sympy: symbolic computing in python. *PeerJ Computer*
593 *Science*, **3**, e103, doi:10.7717/peerj-cs.103.

594 Munk, W. H., 1950: On the wind-driven ocean circulation. *J. Meteorolog.*, **7**, 79–93, doi:10.1175/
595 1520-0469(1950)007⟨0080:otwdoc⟩2.0.co;2.

596 Nilsson, J., J.-A. H. Kallmyr, and P. E. Isachsen, 2024: Topographic steering of the upper Arctic
597 Ocean circulation by deep flows. *Tellus, Ser. A*, **76 (1)**, 206–226, doi:10.16993/tellusa.4072.

598 Nilsson, J., G. Walin, and G. Broström, 2005: Thermohaline circulation induced by bottom friction
599 in sloping-boundary basins. *J. Mar. Res.*, **63 (4)**, 705–728, doi:10.1357/0022240054663222.

600 Nøst, O. A., and P. E. Isachsen, 2003: The large-scale time-mean ocean circulation in the Nordic
601 Seas and Arctic Ocean estimated from simplified dynamics. *J. Mar. Res.*, **61 (2)**, 175–210,
602 doi:10.1357/002224003322005069.

603 Park, Y.-H., and J. M. Guernier, 2001: A simple method for diagnosing the bottom current
604 field of the world's oceans. *J. Phys. Oceanogr.*, **31 (4)**, 972–991, doi:10.1175/1520-0485(2001)
605 031⟨0972:asmfdt⟩2.0.co;2.

606 Pedlosky, J., 1987: *Thermocline theories*, 55–101. Springer-Verlag, doi:10.1007/
607 978-1-4612-4636-7_2.

608 Pedlosky, J., 1996: *Ocean circulation theory*. Springer-Verlag, doi:10.1007/978-3-662-03204-6.

609 Pedlosky, J., 2006: *A History of Thermocline Theory*, 139–152. Springer New York, doi:10.1007/
610 0-387-33152-2_9.

611 Peterson, H. G., and J. Callies, 2023: Coupling between abyssal boundary layers and the interior
612 ocean in the absence of along-slope variations. *J. Phys. Oceanogr.*, **53 (1)**, 307–322, doi:
613 10.1175/jpo-d-22-0082.1.

614 Peterson, H. G., and J. Callies, 2026: Connecting the local response to abyssal mixing to the
615 basin-scale circulation. *J. Phys. Oceanogr.*, **56 (2)**, 431–450, doi:10.1175/JPO-D-25-0159.1.

616 Quarteroni, A., and A. Valli, 2023: *Domain decomposition methods for partial differential equa-*
617 *tions*. Oxford scholarship online, Clarendon Press, Oxford, previously issued in print: 1999.

618 Rhines, P. B., and W. R. Young, 1982: Homogenization of potential vorticity in planetary gyres.
619 *J. Fluid Mech.*, **122**, 347–367, doi:10.1017/s0022112082002250.

620 Robinson, A., and H. Stommel, 1959: The oceanic thermocline and the associated thermohaline
621 circulation. *Tellus A: Dynamic Meteorology and Oceanography*, **11 (3)**, 295–308, doi:10.3402/
622 tellusa.v11i3.9317.

623 Salmon, R., 1998: *Lectures on Geophysical Fluid Dynamics*. Oxford University Press, doi:10.
624 1093/oso/9780195108088.001.0001.

625 Sarkisyan, A. S., and V. M. Ivanov, 1971: The joint effect of baroclinicity and bottom relief as
626 an important factor in the dynamics of ocean currents. *Doklady Akademii Nauk SSSR*, **196**,
627 144–147.

628 Sarmiento, J. L., and K. Bryan, 1982: An ocean transport model for the North Atlantic. *J. Geo-*
629 *phys. Res.*, **87 (C1)**, 394–408, doi:10.1029/jc087ic01p00394.

630 Spall, M. A., 2016: Wind-driven flow over topography. *J. Mar. Res.*, **74 (4)**, 229–248, doi:
631 10.1357/002224016820870657.

632 Stommel, H., 1948: The westward intensification of wind-driven currents. *Eos, Trans. Am. Geo-*
633 *phys. Union*, **29 (2)**, 202–206, doi:10.1029/TR029i002p00202.

634 Stommel, H., 1982: Is the South Pacific helium-3 plume dynamically active? *Earth Planet. Sci.*
635 *Lett.*, **61 (1)**, 63–67, doi:10.1016/0012-821x(82)90038-3.

636 Stommel, H., and A. Arons, 1959: On the abyssal circulation of the world ocean—i. stationary
637 planetary flow patterns on a sphere. *Deep Sea Res.*, **6**, 140–154, doi:10.1016/0146-6313(59)
638 90065-6.

639 Styles, A. F., M. J. Bell, D. P. Marshall, and D. Storkey, 2022: Spurious forces can dominate
640 the vorticity budget of ocean gyres on the C-grid. *J. Advances Model. Earth Sys.*, **14 (5)**,
641 doi:10.1029/2021ms002884.

642 Sverdrup, H. U., 1947: Wind-driven currents in a baroclinic ocean; with application to the
643 equatorial currents of the eastern pacific. *Proc. Nat. Acad. Sci.*, **33 (11)**, 318–326, doi:10.1073/
644 pnas.33.11.318.

645 Timmermans, M., and J. Marshall, 2020: Understanding Arctic Ocean circulation: A review of
646 ocean dynamics in a changing climate. *J. Geophys. Res.*, **125 (4)**, doi:10.1029/2018jc014378.

647 Vallis, G. K., 2006: *Atmospheric and Oceanic Fluid Dynamics: Fundamentals and Large-scale*
648 *Circulation*. 1st ed., Cambridge University Press, Cambridge, United Kingdom and New York,
649 NY, USA, 745 pp., doi:10.1017/cbo9780511790447.

Veronis, G., 1969: On theoretical models of the thermohaline circulation. *Deep Sea Research (Suppl.)*, **16**, 301–323.

Wagner, G. L., and Coauthors, 2025: High-level, high-resolution ocean modeling at all scales with oceananigans. doi:10.48550/ARXIV.2502.14148, 2502.14148.

Waldman, R., and H. Giordani, 2023: Ocean barotropic vorticity balances: Theory and application to numerical models. *J. Advances Model. Earth Sys.*, **15 (4)**, doi:10.1029/2022ms003276.

Welander, P., 1959: An advective model of the ocean thermocline. *Tellus*, **11 (3)**, 309–318, doi:10.1111/j.2153-3490.1959.tb00036.x.

Welander, P., 1968: Wind-driven circulation in one- and two-layer oceans of variable depth. *Tellus*, **20 (1)**, 1–16, doi:10.1111/j.2153-3490.1968.tb00347.x.

Welander, P., 1971: A discussion on ocean currents and their dynamics - the thermocline problem. *Phil. Trans. R. Soc. Lond. A*, **270 (1206)**, 415–421, doi:10.1098/rsta.1971.0081.

Willmott, A. J., and M. V. Luneva, 2015: Steady, barotropic wind and boundary-driven circulation on a polar plane. *Geophys. Astrophys. Fluid Dyn.*, **109 (3)**, 216–233, doi:10.1080/03091929.2015.1050589.

LIST OF TABLES

Table 1. Summary of diagnostic dynamical balances explained in Section 3. The starting point is the three-dimensional momentum (3DM) equations explained in Section 2. The variables are: (u,v,w) velocity, (U,V) depth-integrated horizontal velocity, p pressure, b buoyancy, τ surface stress, p_b bottom pressure, χ baroclinic potential energy, τ_b bottom stress, Ψ streamfunction for depth-integrated horizontal velocity. 44

Diagnostic balance	Equation	Variables	Derivation
3D momentum (3DM)	(11)–(12)	u, v, p, and implicitly τ, b, w	Axiomatic
Depth-integrated momentum (DIM)	(32)–(33)	$U, V, p_b, \chi, \tau, \tau_b$	Integrate 3DM vertically
Barotropic vorticity (BV)	(36)	U, V, p_b, τ, τ_b	Integrate 3DM vertically, then take curl
Depth-integrated vorticity (DIV)	(41)	$V, w\|_{z=-H}, \tau, \tau_b$	Curl of 3DM, then integrate vertically
Geostrophic contour vorticity balance (GCVB)	(43)	Ψ, χ, τ, τ_b	Divide DIM by H, then take curl
Transport divergence vorticity balance (TDVB)	(46)	p_b, χ, τ, τ_b	Divide DIM by f, then take curl

TABLE 1. Summary of diagnostic dynamical balances explained in Section 3. The starting point is the three-dimensional momentum (3DM) equations explained in Section 2. The variables are: (u, v, w) velocity, (U, V) depth-integrated horizontal velocity, p pressure, b buoyancy, τ surface stress, p_b bottom pressure, χ baroclinic potential energy, τ_b bottom stress, Ψ streamfunction for depth-integrated horizontal velocity.

LIST OF FIGURES

Fig. 1. (a) and (c) Example solutions for buoyancy profiles $b(z)$ for various surface values b_s, forcing profiles $B(z)$ (faint colours), and Ekman numbers Ek (see (50)). (b) and (d) show the corresponding Green's functions $G_b(z,\xi)$ for various source points ξ (see (51)). $H = \Gamma = \kappa = 1$ in this figure. 46

Fig. 2. Velocity Green's functions $G_{\mathrm{u}}(z,\xi)$ for various source depths ξ (see (59)). (a) and (b) show the magnitude and phase, respectively, for Ek = 0.25. (c) and (d) are the same, except for Ek = 0.025. $H = f = \nu = 1$ in this figure, and the intensity of the phase colour is proportional to the corresponding magnitude of the Green's function component. The Gu_int quantity is $\int_{-H}^{0} G_{\mathrm{u}}(z;\xi)\, d\xi$. See text for explanation of the phase convention. 47

Fig. 3. Example solution to the barotropic frictional geostrophic flow problem from `Example_barotropic_GeostrophicFlow_v0.2.ipynb`. See also Figure 2.12 from Vallis (2006). 48

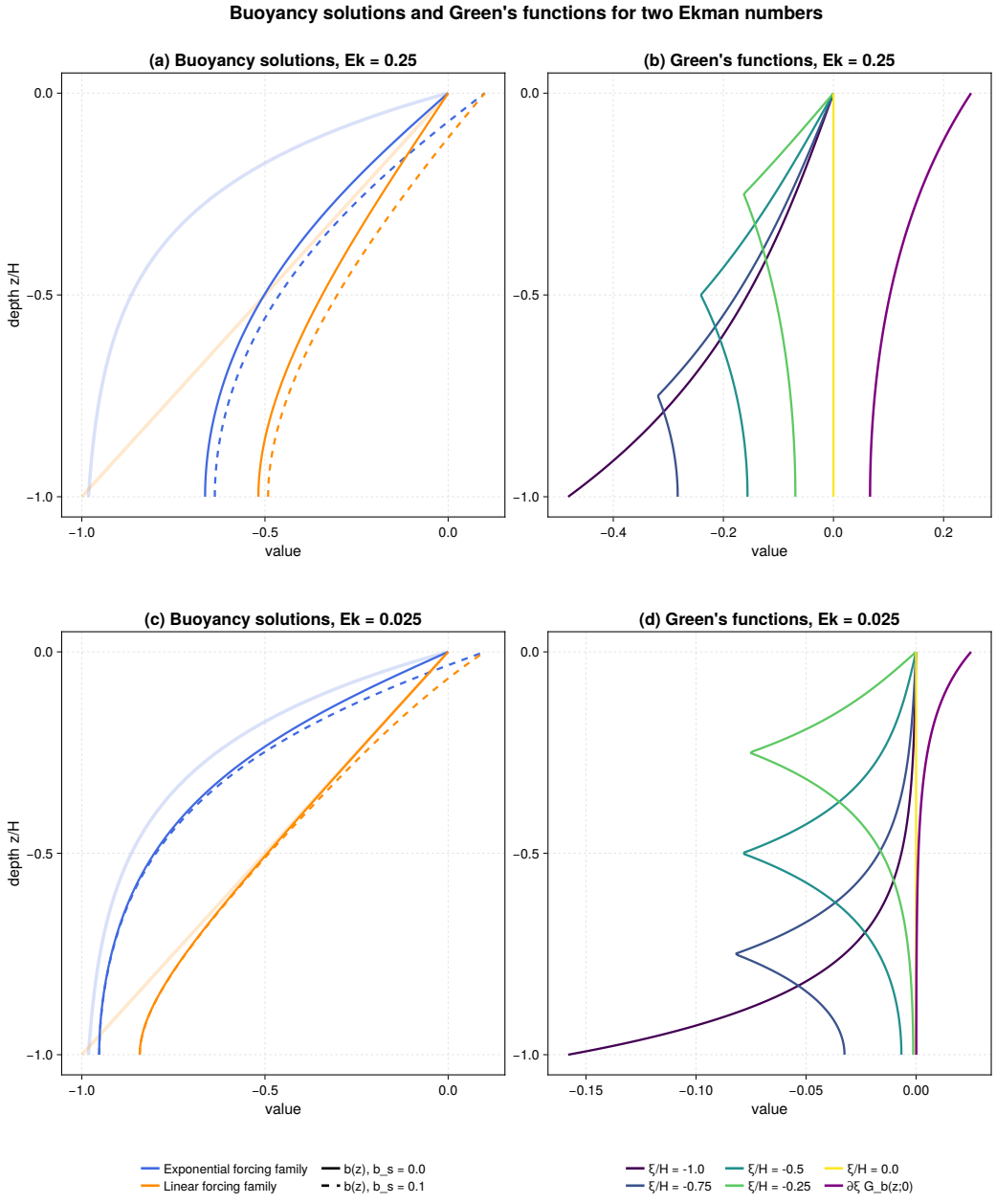
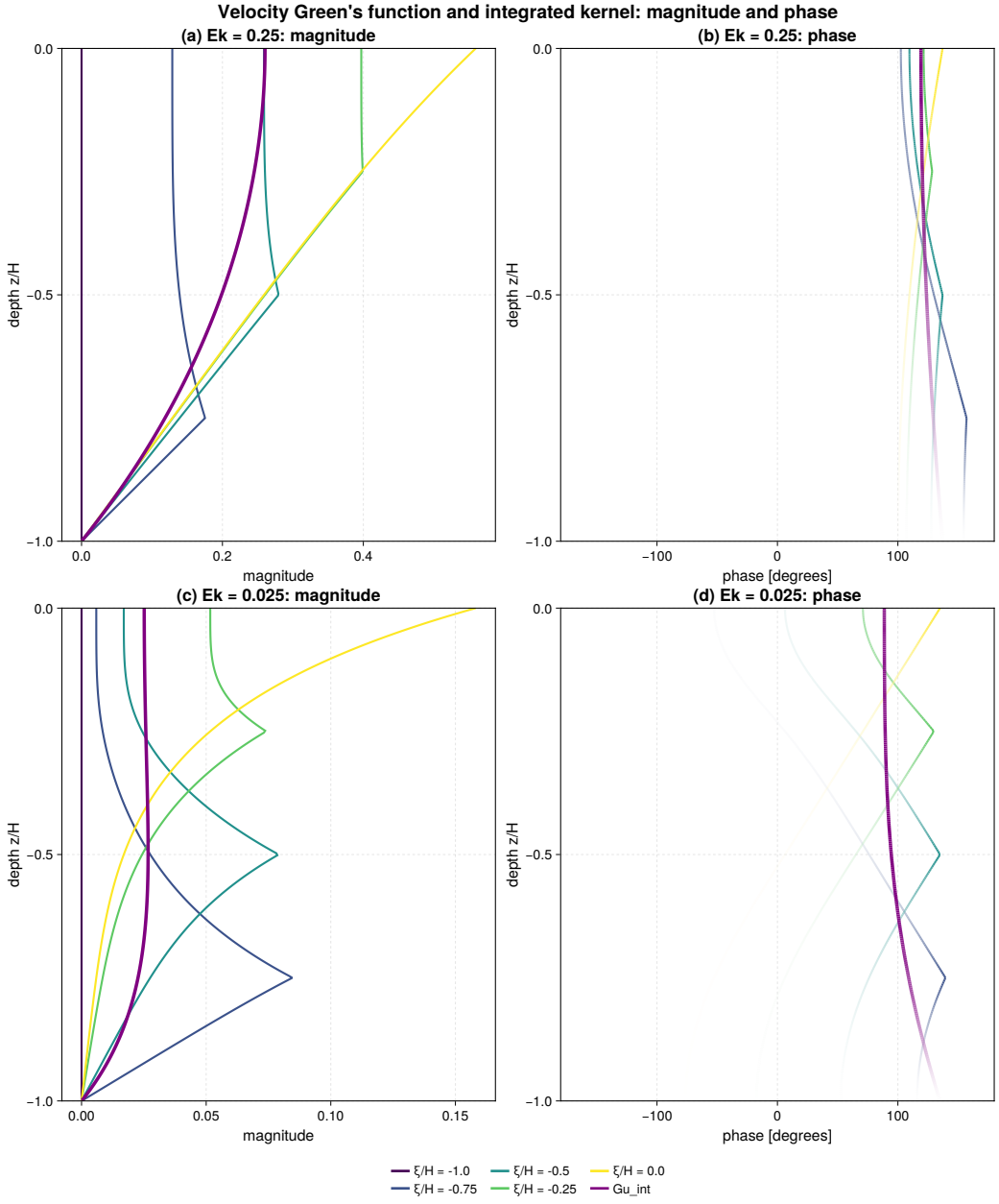


FIG. 1. (a) and (c) Example solutions for buoyancy profiles $b(z)$ for various surface values b_s, forcing profiles $B(z)$ (faint colours), and Ekman numbers Ek (see (50)). (b) and (d) show the corresponding Green's functions $G_b(z,\xi)$ for various source points ξ (see (51)). $H = \Gamma = \kappa = 1$ in this figure.



692 FIG. 2. Velocity Green's functions $G_{\mathrm{u}}(z,\xi)$ for various source depths ξ (see (59)). (a) and (b) show the
693 magnitude and phase, respectively, for Ek = 0.25. (c) and (d) are the same, except for Ek = 0.025. $H = f = \nu = 1$
694 in this figure, and the intensity of the phase colour is proportional to the corresponding magnitude of the Green's
695 function component. The Gu_int quantity is $\int_{-H}^{0} G_{\mathrm{u}}(z;\xi)\, d\xi$. See text for explanation of the phase convention.

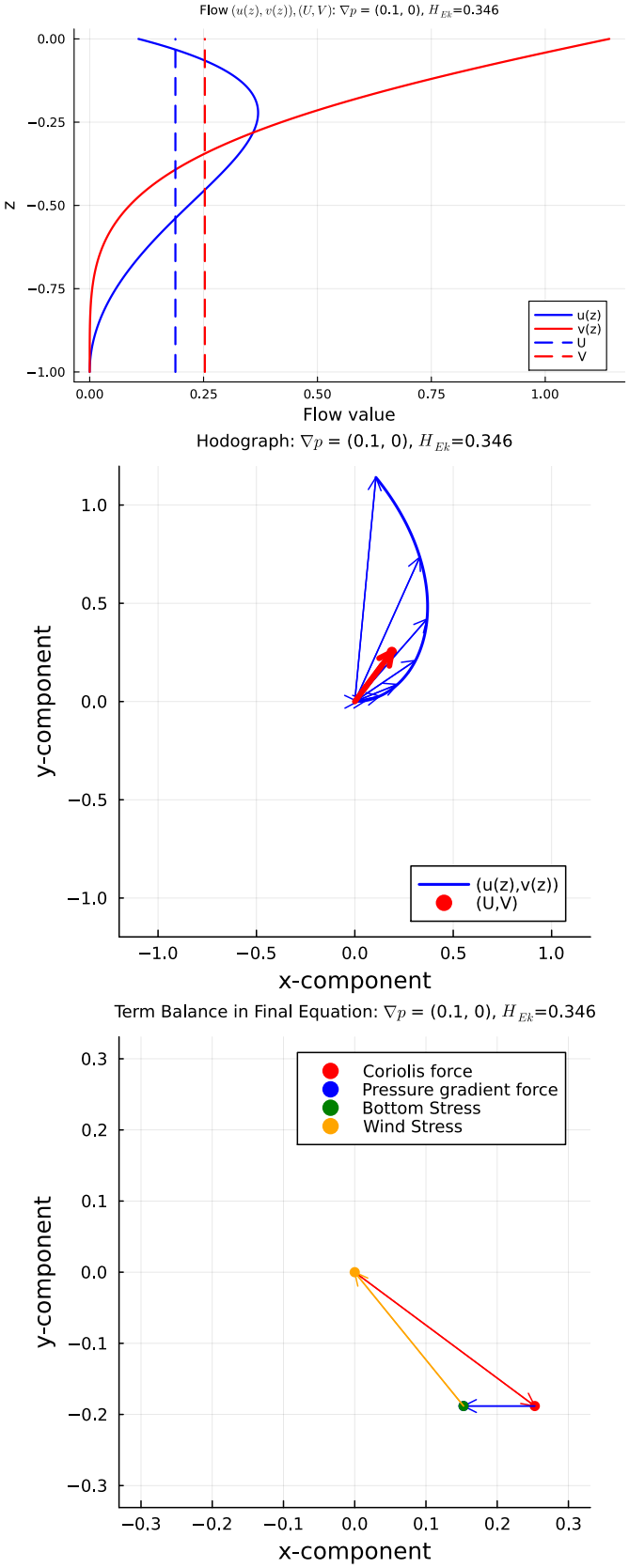


FIG. 3. Example solution to the barotropic frictional geostrophic flow problem from `Example_barotropic_GeostrophicFlow_v0.2.ipynb`. See also Figure 2.12 from Vallis (2006).